

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Data Interpretation

|                  |                                                                                                                                   |               |                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|
| Course Code:     | DSA03                                                                                                                             | Course Title: | Natural Language Processing             |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 1 – Data Interpretation |
| Weightage:       | 35% of CIA                                                                                                                        |               |                                         |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                         |
| Student Name:    | B.VANI                                                                                                                            | Reg. No.:     | 192472348                               |
| Section / Batch: | CSE AI /2024                                                                                                                      | Date:         |                                         |

### Code of Conduct

I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.VANI

### Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

| Section / Topic                                                                | Focus                                                                                                                                                         | Question No. |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language                           | Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems. | Q1           |
| First-Order Predicate Calculus & Representing Linguistically Relevant Concepts | Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.           | Q2           |
| Lexemes and Their Senses & Word Sense Disambiguation                           | Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.                 | Q3           |
| Syntax-Driven Semantic Analysis                                                | Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.           | Q4           |

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

| Customer Query                                                  | Generated Semantic Frame                                      |
|-----------------------------------------------------------------|---------------------------------------------------------------|
| "Transfer ₹20,000 from my savings account to my son's account." | TRANSFER(SourceAccount, Amount, DestinationAccount, Customer) |
| "Block my debit card immediately."                              | BLOCK(DebitCard, Customer)                                    |
| "Send my transaction statement to my email."                    | SEND(TransactionStatement, Email, Customer)                   |
| "Increase my daily withdrawal limit."                           | MODIFY(WithdrawalLimit, Customer)                             |

Additional transaction logs show:

| Query ID | Actual User Intent                     | System Interpretation     |
|----------|----------------------------------------|---------------------------|
| B1       | Transfer money to another account      | Transfer money            |
| B2       | Block debit card                       | Block debit card          |
| B3       | Receive transaction statement by email | Send statement to email   |
| B4       | Increase withdrawal limit              | Decrease withdrawal limit |

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

| Field | Soil Condition | Irrigation Status | Crop  |
|-------|----------------|-------------------|-------|
| F1    | Dry            | Available         | Rice  |
| F2    | Wet            | Available         | Wheat |
| F3    | Dry            | Unavailable       | Maize |
| F4    | Wet            | Unavailable       | Rice  |

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

| Search Query     | Possible Sense 1  | Possible Sense 2          |
|------------------|-------------------|---------------------------|
| "Amazon tour"    | Amazon rainforest | Amazon company            |
| "Safari booking" | Wildlife tour     | Web browser               |
| "Java vacation"  | Java island       | Java programming language |
| "Cruise package" | Ship journey      | Cruise software/project   |

The system records the following user interactions:

| Query          | User Interaction                    |
|----------------|-------------------------------------|
| Amazon tour    | Viewed rainforest trekking packages |
| Safari booking | Selected wildlife resort packages   |
| Java vacation  | Viewed hotels in Bali and Java      |
| Cruise package | Viewed Mediterranean ship packages  |

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

#### Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

#### 4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

| Legal Sentence                                             | Parse Structure      | Generated Semantic Interpretation                       |
|------------------------------------------------------------|----------------------|---------------------------------------------------------|
| "The supplier delivered the equipment to the buyer."       | Subject–Verb–Object  | Supplier = Agent, Equipment = Object, Buyer = Recipient |
| "The buyer received the equipment from the supplier."      | Subject–Verb–Object  | Buyer = Agent, Equipment = Object, Supplier = Recipient |
| "The company terminated the contract after the violation." | Subject–Verb–Object  | Company = Agent, Contract = Object                      |
| "The contract was terminated by the company."              | Passive Construction | Contract = Agent, Company = Object                      |

#### Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

#### Rubrics for Calculation

| Criteria                | Weightage | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                                 | Level 2 (Developing)                     | Level 1 (Beginning)                       |
|-------------------------|-----------|---------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-------------------------------------------|
| Understanding of Data   | 25%       | Accurately interprets all data patterns, trends, and relationships        | Interprets data correctly with minor gaps            | Partial understanding; misses key trends | Misinterprets or fails to understand data |
| Analysis                | 25%       | Provides deep analysis with strong logical reasoning and justification    | Provides reasonable analysis with some justification | Basic analysis with weak reasoning       | No meaningful analysis provided           |
| Application of Concepts | 20%       | Effectively applies relevant concepts/tools to interpret data accurately  | Applies concepts with minor errors                   | Limited application; requires guidance   | Unable to apply concepts to data          |
| Conclusion              | 20%       | Draws clear, meaningful conclusions with strong insights and implications | Provides valid conclusions with moderate insights    | Conclusions are vague or weak            | No clear or valid conclusions             |
| Clarity                 | 10%       | Presents interpretation clearly using proper                              | Generally clear with minor issues                    | Lacks clarity or proper structure        | Poor presentation                         |

|  |  |                                     |  |  |                             |
|--|--|-------------------------------------|--|--|-----------------------------|
|  |  | structure, visuals, and terminology |  |  | and difficult to understand |
|--|--|-------------------------------------|--|--|-----------------------------|

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|-------------------------|-----------------------|----------|-------------------------|------------|---------|------------|---------|
| 1                       |                       |          |                         |            |         |            |         |
| 2                       |                       |          |                         |            |         |            |         |
| 3                       |                       |          |                         |            |         |            |         |
| 4                       |                       |          |                         |            |         |            |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

# DSA0306-NATURAL LANGUAGE PROCESSING

## CO3 AT 2

B.VANI(192472348)

### Q1. FRAME SEMANTICS IN BANKING CUSTOMER-SERVICE SYSTEMS

#### 1. Analyze the semantic frame of each banking request.

The semantic frame identifies the action and its participants.

1. "Transfer ₹20,000 from my savings account to my son's account."  
Semantic Frame:  
TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)  
Action = Transfer  
SourceAccount = Savings account  
Amount = ₹20,000  
DestinationAccount = Son's account  
Customer = User  
Relationship: The customer performs the transfer action, moving the specified amount from the source account to the destination account.
2. "Block my debit card immediately."  
Semantic Frame:  
BLOCK(DebitCard, Customer)  
Action = Block  
Object = Debit card  
Customer = User  
Relationship: The customer requests the system to block the debit card.
3. "Send my transaction statement to my email."  
Semantic Frame:  
SEND(TransactionStatement, Email, Customer)  
Action = Send  
Object = Transaction statement  
Destination = Customer's email  
Customer = User  
Relationship: The customer requests the transaction statement to be sent to the specified email.
4. "Increase my daily withdrawal limit."  
Semantic Frame:  
MODIFY(WithdrawalLimit, Customer)  
Action = Modify/Increase  
Object = Daily withdrawal limit

Customer = User

Relationship: The customer requests an increase in the withdrawal limit.

## **2. Identify the incorrect semantic interpretation.**

The incorrect interpretation is B4.

Query:

"Increase withdrawal limit"

Actual User Intent = Increase withdrawal limit

System Interpretation = Decrease withdrawal limit

The system has reversed the intended action. "Increase" and "decrease" have opposite meanings. Therefore, the semantic interpretation is incorrect.

## **3. How can incorrect semantic roles cause an incorrect banking operation?**

Semantic roles identify who performs an action, what object is affected, the source, destination, and amount involved. If these roles are incorrectly identified, the banking system may perform the wrong operation.

For example, if "Increase withdrawal limit" is interpreted as "Decrease withdrawal limit", the customer's withdrawal limit may be reduced instead of increased. Similarly, confusing the source and destination accounts in a transfer could result in money being transferred from the wrong account to the wrong account. Therefore, incorrect semantic-role assignment can cause financial loss, failed transactions, and serious customer-service errors.

## **4. Recommended frame-based semantic analysis approach.**

A reliable banking conversational system should use the following approach:

Customer Query → Tokenization → Intent Detection → Semantic Frame Extraction  
→ Role Identification → Validation → Banking Operation

The system should identify:

Action → Source → Amount → Destination → Object → Customer

It should also validate critical information before executing a transaction. For example, transfer amount, source account, destination account, and requested operation should be checked against the original query. This frame-based approach reduces ambiguity and prevents incorrect banking operations.

### **Result:**

Frame-based semantic analysis correctly identifies the action and participants in banking requests. The analysis shows that B4 is incorrectly interpreted and that accurate semantic-role identification is essential for safe and reliable banking conversational systems.

## **Q2. FIRST-ORDER PREDICATE CALCULUS FOR SMART AGRICULTURE**

## 1. Represent the field observations using First-Order Predicate Calculus.

The given observations can be represented as:

F1:

Dry(F1)

IrrigationAvailable(F1)

Rice(F1)

F2:

Wet(F2)

IrrigationAvailable(F2)

Wheat(F2)

F3:

Dry(F3)

IrrigationUnavailable(F3)

Maize(F3)

F4:

Wet(F4)

IrrigationUnavailable(F4)

Rice(F4)

Given rules:

$\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$

$\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$

$\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$

$\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$

$\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

For F1:

$\text{Dry}(F1) \wedge \text{IrrigationAvailable}(F1)$

$\rightarrow \text{NeedsWater}(F1)$

$\text{NeedsWater}(F1)$

$\rightarrow \text{Irrigate}(F1)$

$\text{Rice}(F1) \wedge \text{NeedsWater}(F1)$

$\rightarrow \text{PriorityIrrigation}(F1)$

Therefore:

F1 requires irrigation and has priority irrigation.

For F2:

Wet(F2)

$\rightarrow \neg \text{NeedsWater}(F2)$

Therefore:

F2 does not require irrigation.

For F3:

$\text{Dry}(F3) \wedge \text{IrrigationUnavailable}(F3)$

The first rule requires irrigation availability, so  $\text{NeedsWater}(F3)$  cannot be derived from the given rules.

However:

$\text{IrrigationUnavailable}(F3) \rightarrow \neg \text{Irrigate}(F3)$

Therefore:  
F3 should not be irrigated.  
For F4:  
Wet(F4)  
 $\rightarrow \neg \text{NeedsWater}(\text{F4})$   
IrrigationUnavailable(F4)  
 $\rightarrow \neg \text{Irrigate}(\text{F4})$   
Therefore:  
F4 does not require irrigation.

### **Final Result:**

F1  $\rightarrow$  Requires irrigation  
F2  $\rightarrow$  Does not require irrigation  
F3  $\rightarrow$  Should not be irrigated because irrigation is unavailable  
F4  $\rightarrow$  Does not require irrigation

### **3. Should F1 and F3 be treated differently?**

Yes.  
F1 is dry and irrigation is available:  
 $\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1}) \rightarrow \text{NeedsWater}(\text{F1}) \rightarrow \text{Irrigate}(\text{F1})$   
F3 is dry but irrigation is unavailable:  
 $\text{IrrigationUnavailable}(\text{F3}) \rightarrow \neg \text{Irrigate}(\text{F3})$   
Therefore, the controller should irrigate F1 but should not irrigate F3.  
F1 should receive priority because it contains rice and requires water.

### **4. Is predicate logic alone sufficient when sensor data is incomplete or uncertain?**

Predicate logic alone is not sufficient when sensor information is incomplete or uncertain. Traditional predicate logic generally represents facts as true or false. It does not directly express uncertainty such as "the soil is probably dry" or "the sensor is 70% confident".

For agricultural decision support, uncertain sensor readings should be combined with probabilistic or fuzzy approaches. Probabilistic models can represent confidence values, while fuzzy logic can represent gradual conditions such as slightly dry, moderately dry, or very dry.

Therefore, predicate logic is useful for definite rules, but a practical smart agriculture system should combine logic with uncertainty-handling techniques.

### **Result**

First-Order Predicate Calculus successfully represents field conditions and supports rule-based irrigation decisions. F1 requires irrigation, while F3 cannot be irrigated because irrigation is unavailable. However, additional probabilistic or fuzzy methods are required when sensor information is incomplete or uncertain.



### **Q3. WORD SENSE DISAMBIGUATION IN A TRAVEL RECOMMENDATION SYSTEM**

#### **1. Determine the intended sense of the ambiguous terms.**

1. "Amazon tour"  
User interaction: Viewed rainforest trekking packages.  
Intended sense = Amazon rainforest  
Reason: The word "tour" and the user's interaction with trekking packages indicate a travel destination rather than a company.
2. "Safari booking"  
User interaction: Selected wildlife resort packages.  
Intended sense = Wildlife tour  
Reason: "Booking" and "wildlife resort" strongly indicate a safari or wildlife travel experience.
3. "Java vacation"  
User interaction: Viewed hotels in Bali and Java.  
Intended sense = Java island  
Reason: The user interacted with hotels and travel destinations, indicating the Indonesian island rather than the programming language.
4. "Cruise package"  
User interaction: Viewed Mediterranean ship packages.  
Intended sense = Ship journey  
Reason: Mediterranean ship packages clearly indicate a cruise trip.
5. "Safari download for Windows"  
User interaction: Clicked a browser download page.  
Intended sense = Web browser  
Reason: The words "download" and "for Windows", together with the browser download page, indicate the Safari web browser.

#### **2. Identify lexical and contextual cues.**

Amazon:

Lexical cues = "tour", "trekking"

Contextual cue = Viewed rainforest trekking packages

Safari:

Lexical cues = "booking", "wildlife"

Contextual cue = Selected wildlife resort packages

Java:

Lexical cues = "vacation", "hotels"

Contextual cue = Viewed hotels in Bali and Java

Cruise:

Lexical cues = "package", "Mediterranean"

Contextual cue = Viewed ship packages

Safari browser:

Lexical cues = "download", "Windows"

Contextual cue = Clicked browser download page

These clues help distinguish between different lexical senses.

### **3. Why does the same word require different interpretations for different users?**

The meaning of an ambiguous word depends on the surrounding context, user intent, previous searches, and user interactions.

For example:

"Safari booking" → Wildlife tour

"Safari download for Windows" → Web browser

The same word "Safari" has different meanings because the surrounding words and user behaviour provide different contextual evidence. Therefore, WSD must consider more than the word itself.

### **4. Industrial-scale WSD strategy.**

A suitable WSD architecture is:

Query Input → Text Preprocessing → Context Extraction → User History Analysis → Embedding Generation → Candidate Sense Generation → Click Behaviour Analysis → Sense Ranking → Final Sense → Recommendation

The system can use:

1. Query context to analyze nearby words.
2. User history to understand previous interests.
3. Word embeddings or contextual embeddings to measure semantic similarity.
4. Click behaviour to determine which sense the user actually selected.
5. A ranking model to combine all available evidence.

For example:

Sense Score = Query Context + User History + Embedding Similarity + Click Behaviour

The sense with the highest score can be selected.

### **Result**

Word Sense Disambiguation successfully identifies the intended meaning of ambiguous words by combining lexical context, user interactions, and behavioural information. An industrial WSD system should integrate context, history, embeddings, and click behaviour to improve recommendation accuracy.

## **Q4. SYNTAX-DRIVEN SEMANTIC ANALYSIS IN LEGAL DOCUMENT PROCESSING**

### **1. How does syntactic structure influence semantic-role assignment?**

Syntactic structure determines the grammatical relationships between words, but grammatical subject and semantic agent are not always the same.

In active voice:

"The supplier delivered the equipment to the buyer."

The supplier is the grammatical subject and semantic Agent.

The equipment is the Object/Theme.

The buyer is the Recipient.

In:

"The buyer received the equipment from the supplier."

The buyer is the grammatical subject, but semantically the buyer is the Recipient.

The supplier is the Agent because the supplier performed the delivering action.

In passive voice:

"The contract was terminated by the company."

The contract is the grammatical subject, but the semantic Agent is the company. The contract is the Object/Theme.

Therefore, semantic-role assignment must consider syntactic voice and prepositional relationships rather than simply assigning the grammatical subject as the Agent.

### **2. Identify the incorrect semantic-role assignments.**

Sentence 1:

"The supplier delivered the equipment to the buyer."

Given:

Supplier = Agent

Equipment = Object

Buyer = Recipient

This interpretation is correct.

Correct:

Supplier = Agent

Equipment = Object/Theme

Buyer = Recipient

Sentence 2:

"The buyer received the equipment from the supplier."

Given:

Buyer = Agent

Equipment = Object

Supplier = Recipient

This interpretation is incorrect.

Correct:

Buyer = Recipient

Equipment = Object/Theme

Supplier = Agent

Sentence 3:

"The company terminated the contract after the violation."

Given:

Company = Agent

Contract = Object

This interpretation is correct.

Correct:

Company = Agent

Contract = Object/Theme

Sentence 4:

"The contract was terminated by the company."

Given:

Contract = Agent

Company = Object

This interpretation is incorrect.

Correct:

Contract = Object/Theme

Company = Agent

### **Corrected Summary**

| <b>Sentence</b>                                    | <b>Correct Semantic Roles</b>                           |
|----------------------------------------------------|---------------------------------------------------------|
| The supplier delivered the equipment to the buyer. | Supplier = Agent, Equipment = Object, Buyer = Recipient |

The buyer received the equipment from the supplier. Buyer = Recipient, Equipment = Object, Supplier = Agent

The company terminated the contract after the violation. Company = Agent, Contract = Object

The contract was terminated by the company. Contract = Object, Company = Agent

### **3. Effect of confusing grammatical subject with semantic agent.**

A grammatical subject is not always the person or entity performing the action. In active sentences, the subject often acts as the Agent, but in passive constructions the subject can represent the Object or Theme.

For example:

"The contract was terminated by the company."

Grammatical subject = Contract

Semantic Agent = Company

If a legal NLP system assumes that the grammatical subject is always the Agent, it would incorrectly conclude that the contract terminated something. This could result in incorrect extraction of contractual obligations, parties, responsibilities, events, and legal relationships.

Therefore, confusing grammatical and semantic roles can produce serious errors in automated legal information extraction.

### **4. Recommended syntax-driven semantic analysis architecture.**

A suitable architecture is:

Legal Document → Sentence Segmentation → Tokenization → POS Tagging → Dependency/Constituency Parsing → Voice Detection → Semantic Role Labeling → Prepositional Phrase Analysis → Long-Distance Dependency Resolution → Entity Linking → Structured Legal Information

The system should:

1. Detect active and passive voice.
2. Identify subjects, objects, and complements.
3. Analyze prepositional phrases such as "to the buyer" and "from the supplier".
4. Identify semantic roles such as Agent, Object/Theme, and Recipient.
5. Resolve long-distance dependencies across complex legal sentences.

6. Use semantic validation to check whether the resulting interpretation is linguistically and legally consistent.

Example:

"The contract was terminated by the company."

Passive detection:

Contract = grammatical subject

Company = by-phrase

Semantic correction:

Company = Agent

Contract = Object

## **Result**

Syntax-driven semantic analysis correctly combines grammatical structure with semantic roles. The incorrect interpretations in Sentences 2 and 4 arise from confusing grammatical subject with semantic Agent. A system combining parsing, voice detection, semantic-role labeling, and contextual analysis can improve the accuracy of automated legal information extraction.